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Integrated Quantum Technology: Numerical Pulse Optimization

1 Introduction

The development of quantum technologies aims to implement highly precise, robust, and programmable control of quantum bits (qubits). However, physically realizable quantum systems are characterized by their sensitivity to external perturbations, unwanted couplings to the environment, and limitations imposed by decoherence and noise mechanisms, which compromise the temporal stability and fidelity of quantum operations. Within this framework, numerical pulse optimization constitutes a central tool in the design of advanced quantum control strategies.

Pulse optimization is based on formulating the control problem as a functional optimization task, in which one seeks to determine the temporal profiles of the control fields that maximize a performance metric—typically the fidelity of the target unitary operation—subject to the physical constraints of the system. This approach involves the explicit manipulation of the time-dependent Hamiltonian, including the control terms, and the iterative solution of the corresponding evolution equations, such as the time-dependent Schrödinger equation. Compared with analytical techniques or standardized pulse sequences, numerical optimization methods allow realistic experimental effects to be incorporated, such as amplitude limits, finite bandwidth, and parameter mismatches, leading to solutions that can be more robust against systematic imperfections and stochastic fluctuations.

2 Initial Study of the System

In an initial phase, it is necessary to characterize the main physical and operational properties of the quantum system used to carry out the experimental tests. The SPINQ Gemini Lab is a compact quantum computer based on Nuclear Magnetic Resonance (NMR), designed for educational and experimental research purposes.

This system allows the observation, analysis, and manipulation of a quantum register composed of two qubits, with the possibility of extension to a second sample containing three qubits. The system is controlled through the application of radiofrequency pulses, which implement time-dependent control fields and allow controlled rotations to be induced in the system's Hilbert space, enabling the realization of specific unitary operations on the quantum states.

2.1 Larmor Frequency

In the system considered, the qubits are encoded in the nuclear spins of two distinct nuclei: ^1H and ^{31}P . The sample used consists of dimethyl methylphosphonate, whose chemical formula is $(\text{CH}_3\text{O})_2\text{POH}$. Both nuclei have spin quantum number $I = 1/2$, allowing each nucleus to be treated as a two-level quantum system [1, 3].

The computational states are identified with the eigenstates of the spin operator along the quantization axis, such that

$$|0\rangle \equiv |\uparrow\rangle \equiv m_s = +\frac{1}{2}, \quad |1\rangle \equiv |\downarrow\rangle \equiv m_s = -\frac{1}{2}. \quad (1)$$

From a physical standpoint, nuclear spin should not be interpreted as a classical rotation of the nucleus about its own axis, but rather as an intrinsic quantum-mechanical degree of freedom. Each nuclear spin has an associated magnetic moment proportional to the angular momentum operator:

$$\hat{\mu} = \gamma \hat{\mathbf{S}}, \quad (2)$$

where γ represents the gyromagnetic ratio of the nucleus and $\hat{\mathbf{S}} = (S_x, S_y, S_z)$ is the spin angular momentum operator. The presence of this magnetic moment allows the system to couple to external magnetic fields, constituting the fundamental physical mechanism that enables the manipulation and readout of qubits in NMR experiments.

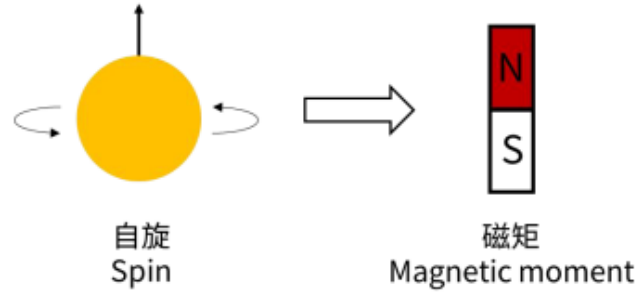


Figure 1: Nuclear spin and its representation in terms of its magnetic moment [3].

When the system is subjected to a static external magnetic field, $\mathbf{B}_0 = B_0 \hat{\mathbf{z}}$, an interaction arises between the nuclear magnetic moment and the applied field. This interaction gives rise to the Zeeman energy, given by

$$E_{\text{mag}} = -\boldsymbol{\mu} \cdot \mathbf{B}_0 = -\mu_z B_0. \quad (3)$$

According to quantum mechanics, the values of the spin angular momentum along the direction of the external field are quantized according to

$$S_z = m_s \hbar \quad \Rightarrow \quad \mu_z = \gamma \hbar m_s. \quad (4)$$

For the nuclei under study, which have spin quantum number $I = 1/2$, the possible values of the magnetic quantum number are

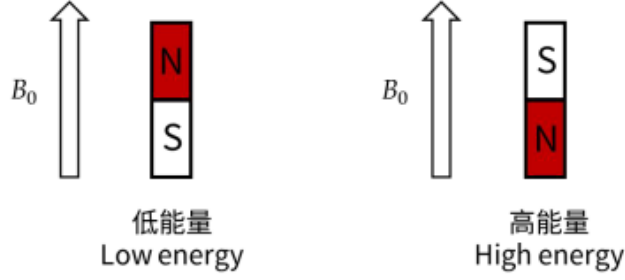


Figure 2: Alignment of the nuclear magnetic moment [3].

$$m_s = \pm \frac{1}{2}. \quad (5)$$

with $m_s = +1/2$ associated with the state $|\uparrow\rangle$ and $m_s = -1/2$ with the state $|\downarrow\rangle$. Thus, the energy levels are given by

$$E = -\gamma\hbar m_s B_0. \quad (6)$$

For $\gamma > 0$, as is the case for both ^1H and ^{31}P , the two energy levels are therefore

$$E_0 = -\frac{\gamma\hbar B_0}{2}, \quad E_1 = +\frac{\gamma\hbar B_0}{2}. \quad (7)$$

The energy difference between the two levels is therefore

$$\Delta E = E_1 - E_0 = \gamma\hbar B_0. \quad (8)$$

When a radiofrequency field is applied with the appropriate resonance frequency, it can induce transitions between these two states. Since $E = hf = \hbar\omega$, the Larmor frequency can be expressed either as an ordinary frequency f_0 or as an angular frequency ω_0 :

$$f_0 = \frac{\Delta E}{h} = \frac{\gamma B_0}{2\pi}, \quad \omega_0 = 2\pi f_0 = \gamma B_0. \quad (9)$$

To describe the coupled two-spin system in the rotating frame, and using frequencies in Hz for consistency with the experimental data, the internal Hamiltonian can be written as

$$\hat{H}_0 = -2\pi f_1 I_{z,1} - 2\pi f_2 I_{z,2} + 2\pi J I_{z,1} I_{z,2}, \quad (10)$$

where $I_{z,i} = \sigma_z^{(i)}/2$, f_1 and f_2 are the resonance frequencies of the two nuclei, and J is the scalar coupling constant between them. In liquids, this scalar coupling is the main mechanism responsible for the interaction between the two spins.

For the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$, the corresponding energy eigenvalues are

$$\begin{aligned} E_{00} &= -\pi f_1 - \pi f_2 + \frac{\pi J}{2}, \\ E_{01} &= -\pi f_1 + \pi f_2 - \frac{\pi J}{2}, \\ E_{10} &= +\pi f_1 - \pi f_2 - \frac{\pi J}{2}, \\ E_{11} &= +\pi f_1 + \pi f_2 + \frac{\pi J}{2}. \end{aligned} \quad (11)$$

The allowed transitions of one spin, while the state of the other spin remains fixed, therefore have frequencies

$$\begin{aligned} \frac{E_{10} - E_{00}}{2\pi} &= f_1 - \frac{J}{2}, & \frac{E_{11} - E_{01}}{2\pi} &= f_1 + \frac{J}{2}, \\ \frac{E_{01} - E_{00}}{2\pi} &= f_2 - \frac{J}{2}, & \frac{E_{11} - E_{10}}{2\pi} &= f_2 + \frac{J}{2}. \end{aligned} \quad (12)$$

Thus, each resonance line is split into two components separated by J in frequency units, or by $2\pi J$ in angular-frequency units.

The transverse component of the nuclear magnetization produces a time-dependent magnetic flux through the detection coil, resulting in an induced electrical signal. The resulting decaying signal is known as the Free Induction Decay (FID). The FID is acquired in the time domain and is commonly transformed into the frequency domain using the Fast Fourier Transform (FFT), allowing the resonance peaks and their splitting to be identified.

2.1.1 Experimental Determination of the Larmor Frequency

Using the SPINQ Gemini Lab and the software dedicated to the analysis of the dimethyl methylphosphonate sample, the first experimental procedure described in the manual [3] consists of determining the intrinsic resonance frequency (Larmor frequency) of each nucleus.

Based on Eq. 9, the tabulated values of the gyromagnetic ratio for each nucleus, and the intensity of the applied external magnetic field ($B_0 \approx 0.65$ T), the resonance frequencies can be estimated as

$$\begin{cases} f_H = \frac{\gamma_H}{2\pi} B_0 = 42.589 \text{ MHz/T} \times 0.65 \text{ T} \approx 27.68 \text{ MHz}, \\ f_P = \frac{\gamma_P}{2\pi} B_0 = 17.258 \text{ MHz/T} \times 0.65 \text{ T} \approx 11.22 \text{ MHz}. \end{cases} \quad (13)$$

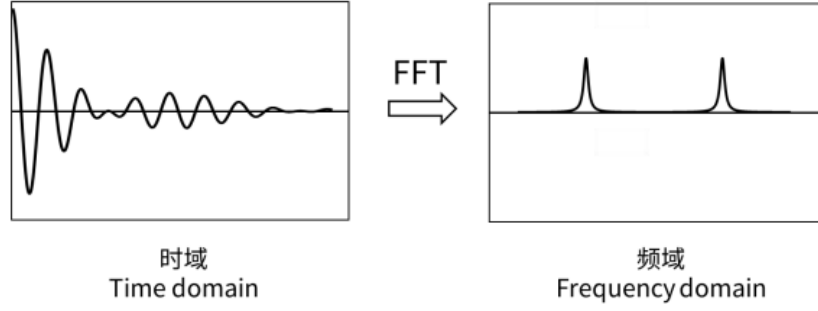


Figure 3: Demonstration of the FFT computed by the computer, allowing visualization of the Larmor frequency [3].

These estimated values were entered into the control software, allowing the system to adjust the radiofrequency field accordingly. However, since the equipment does not operate under strict temperature control, certain physical quantities, particularly the effective magnetic field and consequently the resonance frequencies, are subject to thermal fluctuations and instrumental instabilities. Therefore, the theoretically calculated values do not necessarily guarantee perfect spectral alignment.

To refine the calibration, the spectrum obtained from the FFT of the NMR signal was analyzed. By identifying the midpoint between the spectral peaks corresponding to the two components of each doublet, the frequencies were iteratively adjusted until the midpoint coincided with the origin of the frequency axis.

Once this spectral-centering criterion was satisfied, the adjusted values were used by the system as the experimentally calibrated resonance frequencies.

Under these conditions, it was also possible to determine the scalar coupling constant J from the separation between the doublet peaks observed in the NMR spectrum.

The difference between the frequencies of the two peaks is

$$\Delta f = f_+ - f_-, \quad (14)$$

where f_+ and f_- correspond to the frequencies of the two components of the doublet. With the convention used here, the measured splitting directly provides the coupling constant:

$$J = 695.80 \text{ Hz}. \quad (15)$$

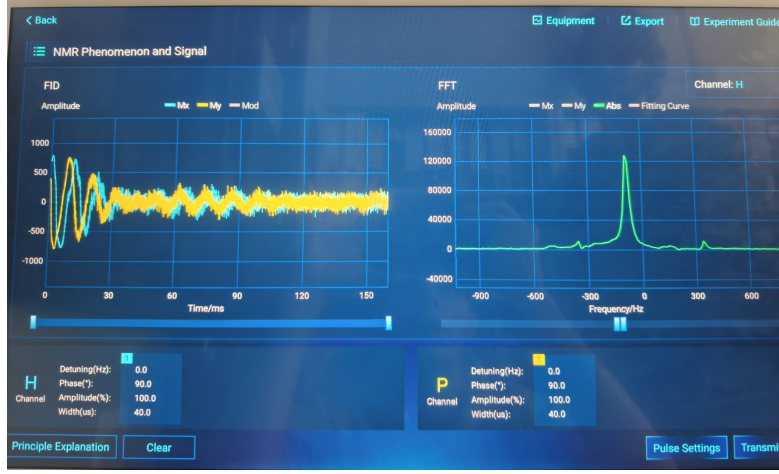


Figure 4: FFT spectrum for ^1H .



Figure 5: FFT spectrum for ^{31}P .

2.2 Rabi Oscillations

When the quantum system is subjected to a radiofrequency (RF) field resonant with the Larmor frequency of a nucleus, coherent transitions between the two logical states of the qubit are induced. These transitions manifest as periodic oscillations between the two levels, with a characteristic frequency known as the Rabi frequency [1, 3].

The RF field applied by the equipment is generated by a coil positioned coaxially with the tube containing the sample. An alternating voltage applied to the coil generates an oscillating magnetic field. In simplified form,

$$V(t) = A_V \cos(2\pi f_{\text{rf}} t + \phi_{\text{rf}}), \quad B_{\text{rf}}(t) = A_{B_{\text{rf}}} \cos(2\pi f_{\text{rf}} t + \phi_{\text{rf}}). \quad (16)$$

The transverse RF field can be represented as

$$\mathbf{B}_{\text{rf}}(t) = A_{B_{\text{rf}}} \cos(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{x}}. \quad (17)$$

Since this is a linearly polarized field, it can be decomposed into two counter-rotating circularly polarized components:

$$\begin{aligned} \mathbf{B}_{\text{rf}}^{(1)} &= B_1 [\cos(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{x}} + \sin(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{y}}], \\ \mathbf{B}_{\text{rf}}^{(2)} &= B_1 [\cos(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{x}} - \sin(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{y}}], \end{aligned} \quad (18)$$

where $B_1 = A_{B_{\text{rf}}}/2$. Within the Rotating Wave Approximation (RWA), only the component rotating in the appropriate direction and close to resonance contributes significantly to the dynamics. The counter-rotating component is highly off-resonant and can therefore be neglected. Thus, the relevant component can be written as

$$\mathbf{B}_{\text{rf}}(t) = B_1 [\cos(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{x}} + \sin(2\pi f_{\text{rf}}t + \phi_{\text{rf}}) \hat{\mathbf{y}}]. \quad (19)$$

Using a rotating reference frame at frequency f_{rf} , the RF field becomes stationary in that frame:

$$\mathbf{B}_{\text{rf}}^{\text{rot}} = B_1 [\cos(\phi_{\text{rf}}) \hat{\mathbf{x}} + \sin(\phi_{\text{rf}}) \hat{\mathbf{y}}]. \quad (20)$$

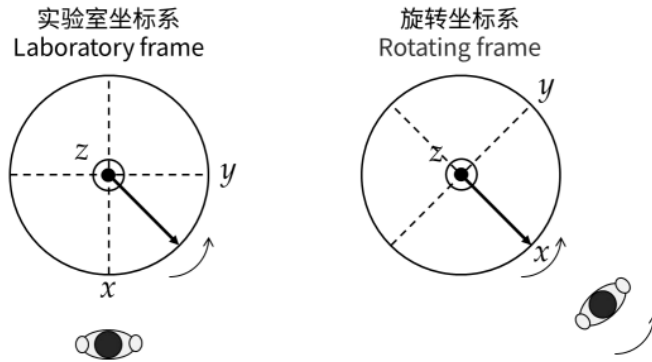


Figure 6: Rotating reference frame [3].

In the rotating frame, the effective longitudinal precession frequency is

$$f_{\text{rot}} = f_0 - f_{\text{rf}}. \quad (21)$$

Under the resonance condition $f_{\text{rf}} = f_0$, the effective longitudinal component vanishes and the spin undergoes rotations about the transverse RF field. The Rabi frequency is then

$$f_1 = \frac{\gamma B_1}{2\pi}. \quad (22)$$

The rotation angle induced by a pulse of duration t_{rf} is therefore

$$\theta = 2\pi f_1 t_{\text{rf}} \quad \Rightarrow \quad t_{\text{rf}} = \frac{\theta}{2\pi f_1}. \quad (23)$$

2.2.1 Experimental Determination of Rabi Oscillations

The second experimental procedure described in the manual [3] consists of applying a sequence of pulses along the y axis, with the aim of inducing rotations of the macroscopic magnetic moment in Bloch space and transferring the magnetization to the transverse component.

After preparing the state magnetized along x , the experimental data are fitted to a sinusoidal function of the form

$$A \sin(2\pi f t). \quad (24)$$

where $f = f_1$ corresponds to the Rabi frequency associated with the amplitude of the applied radiofrequency field.

The fit performed on the experimental data for the ^1H nucleus yielded a quality coefficient of 99.86%, with the following estimated frequency:

$$f_1^H = 6257.66 \text{ Hz} \quad \Rightarrow \quad \frac{1}{f_1^H} = 159.8 \mu\text{s}$$

Similarly, the fit corresponding to the ^{31}P nucleus yielded a quality coefficient of 99.82%:

$$f_1^P = 6187.98 \text{ Hz} \quad \Rightarrow \quad \frac{1}{f_1^P} = 161.6 \mu\text{s}$$

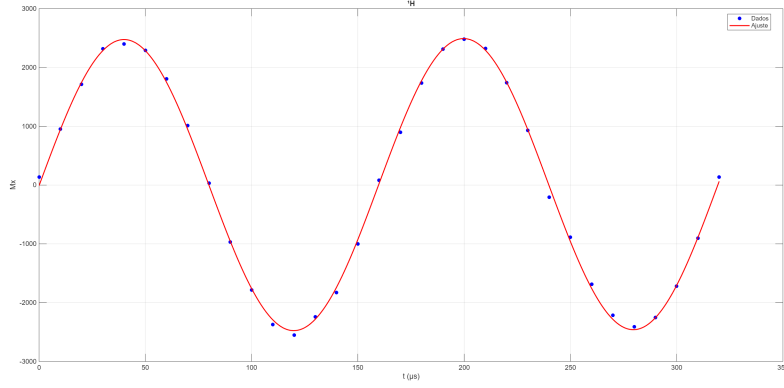


Figure 7: Fit of the experimental data for the ^1H nucleus, performed in MATLAB.

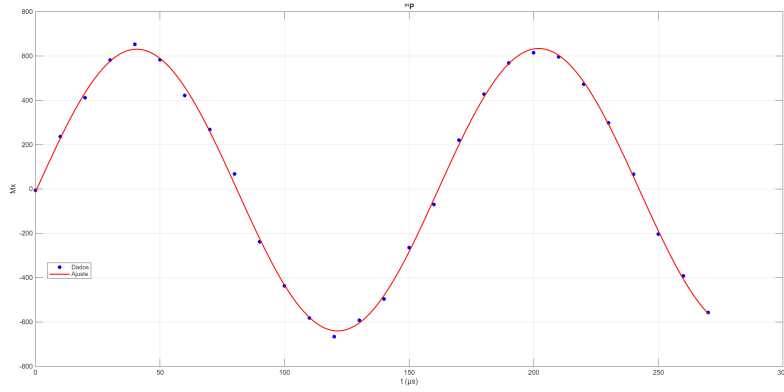


Figure 8: Fit of the experimental data for the ^{31}P nucleus, performed in MATLAB.

2.3 Quantum Decoherence

As discussed previously, quantum systems based on nuclear spin control rely on various physical mechanisms that allow the fundamental states forming the basis of quantum computing to be prepared and manipulated. However, despite the high precision achieved in the control processes, qubits cannot indefinitely preserve a prepared quantum state [1, 3].

Due to unavoidable coupling to the environment, external noise, and experimental imperfections, the system undergoes quantum decoherence, characterized by the progressive loss of phase coherence and degradation of quantum superpositions. This process leads to the loss of quantum information and imposes fundamental limits on operation times and on the fidelity of implemented logic gates.

Decoherence in NMR systems can be experimentally characterized through the longitudinal and transverse relaxation times, T_1 and T_2 , respectively.

Longitudinal relaxation, characterized by T_1 , describes the recovery of the longitudinal magnetization toward its thermal-equilibrium value. After the application of an RF pulse, the longitudinal component of the macroscopic magnetization follows

$$M_z(t) = M_z^{\text{eq}} + [M_z(0) - M_z^{\text{eq}}] e^{-t/T_1}. \quad (25)$$

Table 1: RF pulse durations required to implement rotations of a defined angle on the Bloch sphere.

θ	$t_{\text{rf}}(\theta)$	t_1^H (μs)	t_1^P (μs)
$\pi/4$	$1/(8f_1)$	19.97	20.2
$\pi/2$	$1/(4f_1)$	39.95	40.4
$3\pi/4$	$3/(8f_1)$	59.93	60.6
π	$1/(2f_1)$	79.90	80.8

This process, also known as spin-lattice relaxation, determines the timescale over which the system recovers its equilibrium polarization along the direction of the static magnetic field B_0 . In quantum-computing terms, T_1 is associated with population relaxation and dissipative errors.

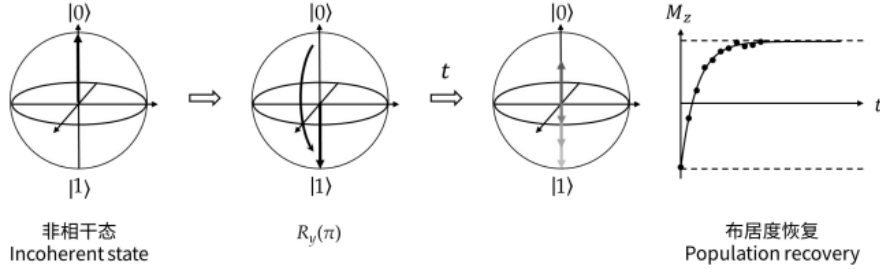


Figure 9: Representation of the process used to visualize longitudinal relaxation [3].

Transverse relaxation, characterized by T_2 , describes the loss of phase coherence. After a pulse that generates magnetization in the transverse plane, the transverse component decays approximately as

$$M_{xy}(t) = M_{xy}(0)e^{-t/T_2}. \quad (26)$$

This phenomenon, commonly referred to as spin-spin relaxation, is associated with fluctuations in the local magnetic environment and interactions between spins. In the density-matrix formalism, T_2 is related to the decay of the off-diagonal terms, which contain the phase-coherence information.

In general,

$$T_2 \leq T_1. \quad (27)$$

In experimental practice, the time T_2^* is frequently measured. Unlike T_2 , it also includes the effects of static magnetic-field inhomogeneity and other quasi-static frequency offsets.

In the context of NMR-based quantum computing, T_1 and T_2 impose fundamental limits on the number of logic operations that can be performed with high fidelity before the quantum information is degraded. Consequently, pulse optimization techniques are essential for reducing the impact of these limitations and improving the fidelity of quantum operations.

2.3.1 Experimental Determination of T_1 and T_2

The third experimental procedure explores the coherent manipulation of quantum states through control of the radiofrequency pulse duration, with the aim of experimentally determining the relaxation times T_1 and T_2 [3].

To measure the longitudinal relaxation time T_1 , the system is initially prepared in an inverted population state by applying an $R_y(\pi)$ pulse. After this inversion, a free-evolution interval is applied during which the system evolves under the internal Hamiltonian and relaxation mechanisms. The longitudinal magnetization is then monitored as a function of the waiting time and fitted to the characteristic exponential recovery associated with T_1 .

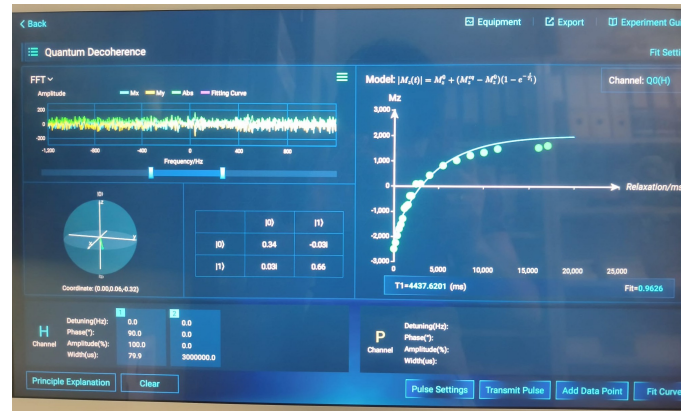


Figure 10: Fit performed by the SPINQ software to determine T_1^H .



Figure 11: Fit performed by the SPINQ software to determine T_1^P .

The measured T_1 times were

$$\begin{aligned} T_1^H &= 4.437 \text{ s,} \\ T_1^P &= 11.479 \text{ s.} \end{aligned}$$

The fit quality was 96.26% for hydrogen and 98.27% for phosphorus.

To determine the transverse relaxation time T_2 , a spin-echo sequence is used. Initially, an $R_y(\pi/2)$ pulse is applied, placing the magnetization in the transverse plane. This is followed by a free-evolution period during which dephasing occurs. At the midpoint of the evolution interval, an $R_x(\pi)$ pulse is applied, reversing the effects of static frequency offsets and refocusing the magnetization. The echo amplitude is then measured and its decay is used to estimate T_2 .

This procedure reduces the contribution of static field inhomogeneity and therefore provides a more accurate estimate of the intrinsic transverse relaxation time.

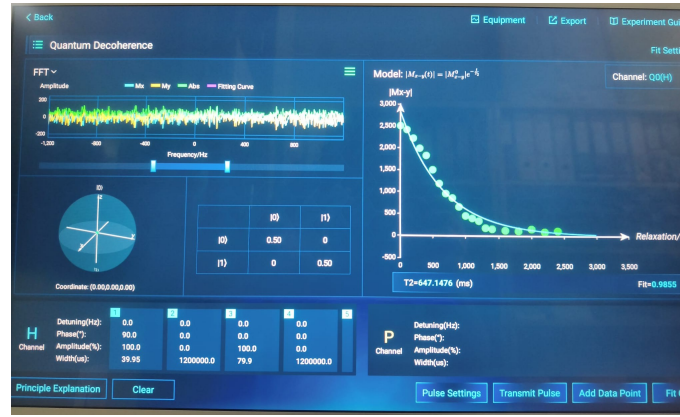


Figure 12: Fit performed by the SPINQ software to determine T_2^H .

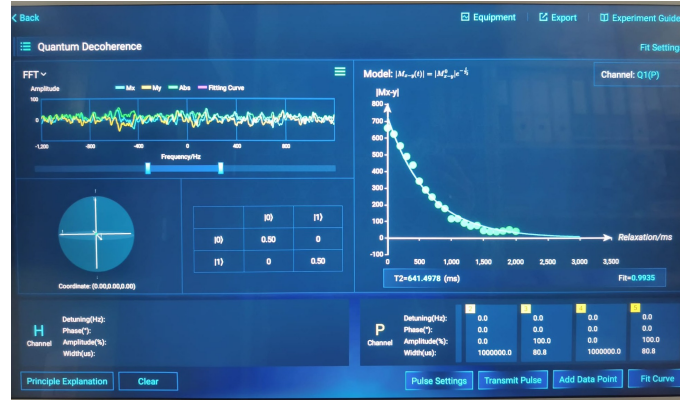


Figure 13: Fit performed by the SPINQ software to determine T_2^P .

The measured T_2 times were

$$\begin{aligned} T_2^H &= 0.647 \text{ s}, \\ T_2^P &= 0.642 \text{ s}. \end{aligned}$$

The fit quality was 98.55% for hydrogen and 99.35% for phosphorus.

3 Study of Numerical Pulse Optimization

Having established the physical characterization of the system and described the fundamental experimental procedures, it is now possible to introduce the numerical pulse optimization approach.

The control mechanisms presented previously are based on applying radiofrequency pulses to the sample nuclei, which induce controlled rotations in the state space and allow the qubits to be prepared, manipulated, and subsequently measured. Formally, these pulses correspond to the introduction of time-dependent control terms into the system Hamiltonian [3].

The total Hamiltonian can be written as

$$H(t) = H_0 + H_1(t), \quad (28)$$

where H_0 represents the internal Hamiltonian of the quantum system and $H_1(t)$ represents the time-dependent control Hamiltonian generated by the RF fields.

For a weakly coupled two-spin system, the internal Hamiltonian in the rotating frame can be expressed as

$$H_0 = 2\pi \sum_i \Delta f_i I_z^{(i)} + 2\pi \sum_{i < j} J_{ij} I_z^{(i)} I_z^{(j)}, \quad (29)$$

where Δf_i represents the frequency offset of spin i relative to the rotating frame and J_{ij} is the scalar coupling constant.

The Pauli matrices are

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (30)$$

with $I_\alpha = \sigma_\alpha/2$.

The control Hamiltonian can then be written as

$$H_1(t) = 2\pi \sum_i f_1^{(i)} \left[\cos(\phi_i(t)) I_x^{(i)} + \sin(\phi_i(t)) I_y^{(i)} \right], \quad (31)$$

where $f_1^{(i)}$ is the Rabi frequency associated with spin i and $\phi_i(t)$ is the phase of the corresponding RF field.

The dynamics of the system is governed by the time-evolution operator

$$U(t) = \mathcal{T} \exp \left[-i \int_0^t H(t') dt' \right], \quad (32)$$

where $\mathcal{T}$ denotes time ordering. If the Hamiltonian is constant during a given time segment, the corresponding propagator reduces to

$$U_k = e^{-iH_k t_k}. \quad (33)$$

It is precisely within this framework that numerical pulse optimization becomes relevant: instead of relying exclusively on standardized elementary sequences, the aim is to determine temporal control profiles that maximize the fidelity of a target operation while accounting for the physical limitations and decoherence mechanisms of the system.

3.1 Hard Pulses

Hard pulses are characterized by relatively high amplitude and short duration compared with the characteristic timescale associated with the internal Hamiltonian. In the weak-coupling approximation, the pulse duration should satisfy approximately

$$t_p \ll \frac{1}{J}, \quad (34)$$

so that the effect of the scalar coupling during the pulse can be neglected to first order.

Under these conditions, the evolution during the pulse is dominated by the control Hamiltonian, allowing approximately ideal rotations to be implemented.

It should be noted that hard pulses are broadband rather than intrinsically selective. Their short duration corresponds to a broad excitation bandwidth, allowing them to act on spins whose resonance frequencies lie within the excitation bandwidth of the pulse.

3.2 Strongly Modulated Pulses

Strongly Modulated Pulses (SMP) constitute a numerical pulse-optimization technique in which the total pulse sequence is divided into a relatively small number of segments. Each segment is characterized by adjustable parameters such as

- amplitude (A),
- phase (ϕ),

- frequency (f_{rf}),
- duration (t).

These parameters are numerically optimized so that the total operation

$$U_{\text{SMP}} = \prod_{k=N}^1 e^{-iH_k t_k} \quad (35)$$

approximates the desired target unitary U_{target} with high fidelity. The unitary fidelity used in this work is

$$F = \frac{|\text{Tr}(U_{\text{target}}^\dagger U_{\text{SMP}})|}{2^n}, \quad (36)$$

where n is the number of qubits and 2^n is the dimension of the Hilbert space. This fidelity ranges from 0 to 1, with $F = 1$ indicating perfect agreement between the two unitary operations up to a global phase.

The cost function is consequently defined as

$$Q = 1 - F, \quad (37)$$

so that the optimization problem consists of minimizing Q .

In the implementation used in this work, the nonlinear optimization was performed using MATLAB's *fminunc* function. Since *fminunc* is designed for unconstrained optimization, the physical limits of the control parameters were incorporated into the parameterization or initial-value generation used by the optimization procedure.

3.3 GRAPE

The GRAPE (Gradient Ascent Pulse Engineering) method is a numerical optimal-control technique in which the control pulse is discretized into a relatively large number of fixed-duration time segments. Within each segment, the Hamiltonian is assumed to be constant, allowing the total propagator to be expressed as an ordered product of elementary propagators.

The main advantage of GRAPE is the efficient calculation of the gradient of the fidelity or cost function with respect to the control amplitudes. This avoids the need to estimate the gradient solely through repeated finite-difference evaluations and allows a large number of control parameters to be optimized efficiently.

In this work, the pulse was divided into $N = 300$ segments, providing a substantially finer temporal discretization than that used in the SMP approach.

3.4 Experimental Use of the SMP and GRAPE Methods

MATLAB was used as the simulation and numerical-computation platform. The initial pulse parameters, namely the amplitudes, phases, and/or durations of the pulse segments, were initialized according to the physical limits of the experimental system.

Different initial conditions were tested in order to explore different regions of the parameter space and reduce the dependence of the optimization on a single initial guess.

3.4.1 Analysis of Hard Pulses

The experimental procedure begins with the explicit definition of the target matrix, corresponding to a unitary operation that is to be implemented in the system. The target operator satisfies the unitarity condition

$$U^\dagger U = I. \quad (38)$$

This condition guarantees preservation of the norm of the quantum state under the ideal unitary evolution.

The target operation considered in the experiment is

$$U = R_{-z}^2(\pi/4)R_{-y}^2(\pi/4)R_z^1(\pi/4)R_y^1(\pi/4). \quad (39)$$

Each rotation is characterized by its rotation axis, rotation angle, and the nucleus on which the operation acts.

In general, a single-qubit rotation is represented by

$$R_\alpha^i(\theta) = e^{-i\frac{\theta}{2}\sigma_\alpha^i} = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)\sigma_\alpha^i. \quad (40)$$

For ^1H ,

$$U^1 = R_z^1(\pi/4)R_y^1(\pi/4) = \begin{pmatrix} 0.8536 - 0.3536i & -0.3536 + 0.1464i \\ 0.3536 + 0.1464i & 0.8536 + 0.3536i \end{pmatrix}. \quad (41)$$

For ^{31}P ,

$$U^2 = R_{-z}^2(\pi/4)R_{-y}^2(\pi/4) = \begin{pmatrix} 0.8536 + 0.3536i & 0.3536 + 0.1464i \\ -0.3536 + 0.1464i & 0.8536 - 0.3536i \end{pmatrix}. \quad (42)$$

The total two-qubit operation is then constructed using the tensor product,

$$U = U^1 \otimes U^2. \quad (43)$$

The initial quantum state is

$$|\psi(0)\rangle = |00\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}. \quad (44)$$

The ideal final state is therefore

$$|\psi(t)\rangle = U|\psi(0)\rangle. \quad (45)$$

For a pure state, the corresponding density matrix is

$$\rho = |\psi(t)\rangle\langle\psi(t)|. \quad (46)$$

The resulting theoretical density matrix was calculated in MATLAB.

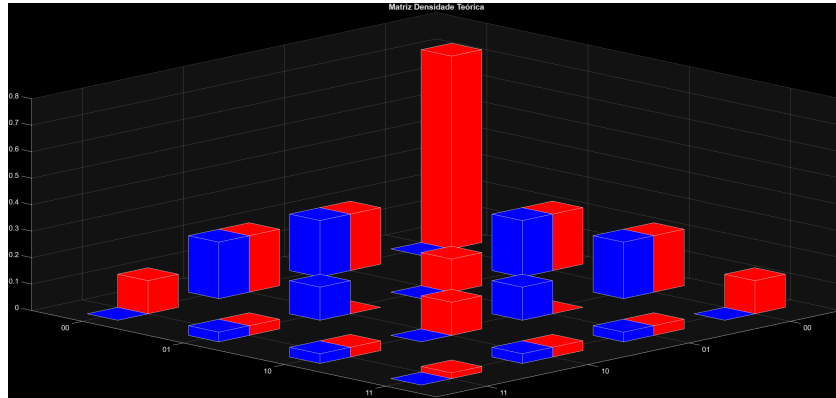


Figure 14: Theoretical density matrix calculated using MATLAB.

After calculating U_{target} , the experimental procedure proceeds with the definition of the pulse durations required to implement the rotations. Based on the previously determined Rabi frequencies, each rotation was parameterized according to its axis and angle.

Rotations about the z axis were not implemented directly because the applied RF pulses act in the transverse plane. Instead, the required z rotations were decomposed into rotations about transverse axes according to

$$R_z(\theta) = R_y(\pi/2)R_x(\theta)R_y(-\pi/2). \quad (47)$$

The pulses were introduced into the system through a configuration file in JSON (JavaScript Object Notation) format and repeated five times before the final data were exported for analysis.

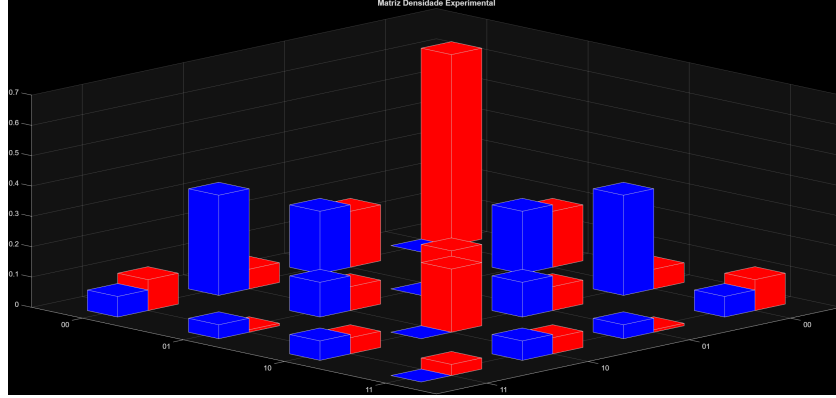


Figure 15: Density matrix generated from the experimental data obtained using hard pulses.

Since the experimental unitary operator $U_{\text{experimental}}$ cannot be directly accessed, the fidelity was calculated between the theoretical and experimentally reconstructed density matrices. Using the convention adopted in this work,

$$F(\rho, \rho_{\text{exp}}) = \text{Tr} \left[\sqrt{\sqrt{\rho_{\text{exp}}} \rho \sqrt{\rho_{\text{exp}}}} \right]. \quad (48)$$

The fidelity calculated from the experimental data obtained using hard pulses was

$$F_{\text{HP}} = 90.04\%.$$

Next, the SMP algorithm was implemented in MATLAB. The procedure included:

- initialization of the system parameters, including the scalar coupling constant J , the maximum Rabi frequency, the Pauli matrices, the target unitary matrix, and the number of pulse segments ($N = 6$);
- generation of different initial parameter sets within the physical operating range of the system;
- evaluation of the cost function using the function *cost_fun()*;
- extraction of the optimized parameters using *extr_p()*;
- generation of the pulse input file in JSON format using *generate_json_file()*;

- minimization of the cost function using MATLAB's *fminunc* optimization routine.

The resulting pulse file was generated and loaded into the quantum computer.

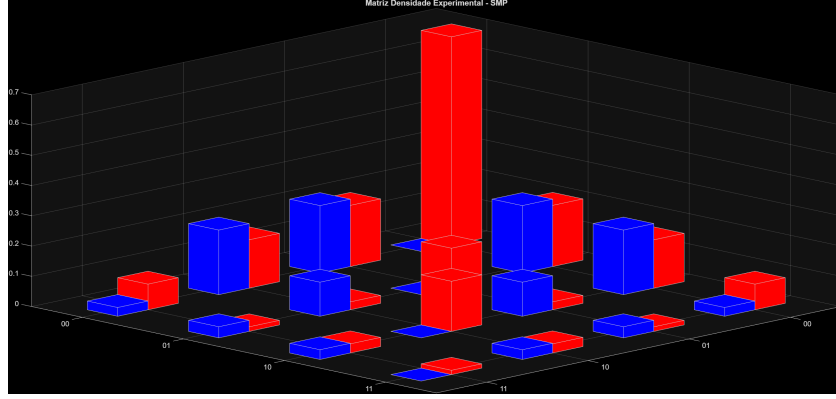


Figure 16: Density matrix generated from the experimental data obtained using the SMP-optimized pulses.

The fidelity calculated for the pulses generated by SMP was

$$F_{\text{SMP}} = 96.82\%.$$

For the application of the GRAPE algorithm, the target operation was redefined with the aim of preparing the Bell state

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle). \quad (49)$$

A unitary operator whose first column corresponds to this target state was selected as

$$U = \begin{pmatrix} 0.7071 & 0 & 0 & 0.7071 \\ 0 & 0.7071 & 0.7071 & 0 \\ 0 & 0.7071 & -0.7071 & 0 \\ 0.7071 & 0 & 0 & -0.7071 \end{pmatrix}. \quad (50)$$

When this operation is applied to the initial state $|00\rangle$, the resulting state is the desired Bell state, up to the numerical precision of the matrix elements.

The GRAPE algorithm was subsequently applied to determine the control profiles capable of implementing the target operation. In contrast with the SMP approach, the pulse was discretized into a substantially larger number of fixed-duration segments, with $N = 300$ segments used in this work.

The resulting fidelity was

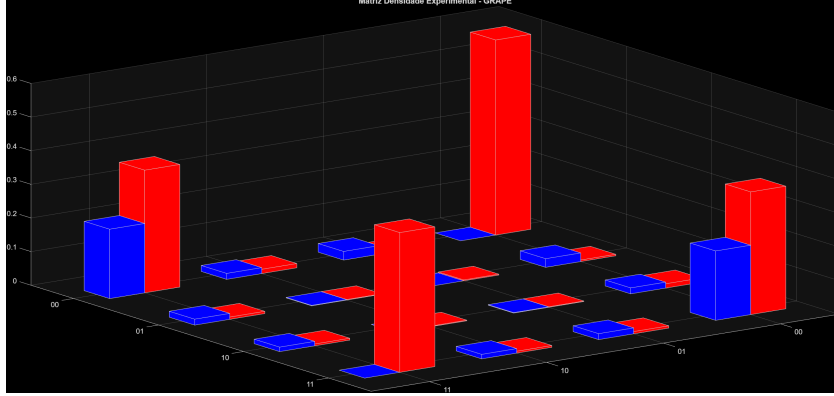


Figure 17: Density matrix generated from the experimental data obtained using GRAPE to prepare the Bell state $|\beta_{00}\rangle$.

$$F_{\text{GRAPE}} = 92.86\%$$

4 Conclusion

The experimental results demonstrate that satisfactory fidelities can be obtained using hard pulses, which provide a simple and effective method for implementing single-qubit rotations when the pulse duration is sufficiently short compared with the characteristic timescales of the internal Hamiltonian. However, experimental imperfections, including calibration drifts, magnetic-field inhomogeneities, frequency offsets, and decoherence, can reduce the fidelity of the resulting operations.

Numerical optimization provided an improvement in the fidelity for the SMP implementation, increasing the measured fidelity from 90.04% for the hard-pulse implementation to 96.82% for the SMP-optimized pulse. This result illustrates the ability of numerical optimization to compensate, at least partially, for imperfections in the experimental implementation.

For the GRAPE experiment, a fidelity of 92.86% was obtained for the preparation of the target Bell state. The direct comparison between the SMP and GRAPE fidelities should, however, be interpreted with caution because the two methods were applied to different target operations and experimental pulse configurations. Therefore, the numerical values cannot be interpreted as a direct ranking of the two optimization methods.

The potential of both techniques is also demonstrated in the literature. T. S. Mahesh and D. Suter [2] reported SMP fidelities above 97.9% for a strongly coupled four-spin system. Similarly, Xiao-dong Yang *et al.* [4] reported GRAPE fidelities above 99% under their experimental conditions.

The differences between these reported values and those obtained in the present experiment can be associated with differences in the physical systems, pulse parameterizations, calibration procedures, environmental stability, and experimental conditions. In particular, temperature fluctuations, magnetic-field inhomogeneities, imperfect knowledge of system parameters, and relaxation mechanisms can limit the experimentally achievable fidelity.

Overall, the results demonstrate the relevance of numerical pulse optimization for quantum control. The experimental implementation of SMP and GRAPE provides a practical illustration of how numerical methods can be used to design control pulses that account for the dynamics of a real quantum system and improve the fidelity of experimentally implemented quantum operations.

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